

David G Courtney BSc PhD

Senior Lecturer in Molecular Virology, Queen's University Belfast
UK Ambassador for the European Research Council

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Molecular virologist and RNA biologist. David leads The Courtney Group at Queen's University Belfast, which studies how influenza A virus RNA recruits host machinery to be modified, exported, translated and trafficked, and how those dependencies can be turned into RNA-based therapeutics. He holds an ERC Starting Grant and represents the United Kingdom as an Ambassador for the European Research Council.

1,531 citations · h-index 19 · 52 indexed works (OpenAlex, 2026-10-07)

APPOINTMENTS

2025 –	Senior Lecturer in Molecular Virology , Queen's University Belfast, UK
2021 – 2025	ERC Research Fellow in Molecular Virology , Queen's University Belfast, UK Founded The Courtney Group with an ERC Starting Grant.
2020 – 2021	Postdoctoral Research Fellow in Molecular Virology , Queen's University Belfast, UK High-throughput drug repurposing screens against SARS-CoV-2 in airway epithelial cells.
2019 – 2020	Marie Skłodowska-Curie Fellow, Virology and Epitranscriptomics , Institut Pasteur, Paris, France Developed methods to identify the host RNA-binding proteins of influenza A virus RNAs.
2016 – 2019	Postdoctoral Research Fellow, Virology and Epitranscriptomics , Duke University, North Carolina, USA Showed how m6A and m5C RNA modifications regulate influenza A virus, HIV-1 and other viruses.
2015	Postdoctoral Research Associate, CRISPR gene editing , Ulster University, UK Allele-specific CRISPR/Cas9 therapy for an inherited corneal dystrophy, tested in vivo.
2010 – 2011	Industrial placement, BioPharm R&D , GlaxoSmithKline, UK Twelve months in molecular expression and cell line development.

EDUCATION

2012 – 2015	PhD, Molecular Biology , Ulster University, UK The development of personalised medicines for corneal dystrophies (siRNA and CRISPR therapeutics).
2008 – 2012	BSc (Hons) Biomedical Sciences with industrial studies, First Class , Ulster University, UK

LEADERSHIP AND SERVICE

2025 –	UK Ambassador for the European Research Council , Ambassadors for the ERC network One of 32 ERC grantees chosen to make the case for frontier research to policymakers, the media and the research community.
Current	STEM NI Ambassador , Northern Ireland
2014 –	Fellow of the Winston Churchill Memorial Trust , Churchill Fellowship, UK

GRANTS AND AWARDS

2020	ERC Starting Grant , European Research Council €1,568,010 over five years to establish The Courtney Group and study the regulation of influenza A virus RNAs.
2017	Marie Skłodowska-Curie Global Fellowship , European Commission €260,000 for three years of research at Duke University and Institut Pasteur.
2015	Postgraduate Student of the Year (John RE Scott Convocation award) , Ulster University
2015	EMBO Travel Grant , EMBO
2014	Churchill Fellowship , Winston Churchill Memorial Trust Research visits to Modena, Italy and Aarhus, Denmark.
2013	Young Investigator Award, best oral presentation , Irish Society of Human Genetics

PATENT APPLICATIONS

- 2019 **Compositions and methods for enhanced gene expression and viral replication**, US patent application 16/091,822
Using m6A-associated cellular machinery to boost transgene expression or viral replication.
- 2019 **Single guide RNA/CRISPR/Cas9 systems, and methods of use thereof**, US patent application 16/326,908
Allele-specific CRISPR/Cas9 systems for treating corneal dystrophies.

PROFESSIONAL MEMBERSHIPS

Microbiology Society · American Society for Virology · RNA Society

SELECTED PUBLICATIONS

co-senior author · * co-first author. Full list at davidgcourtney.com/publications and on ORCID.

1. Bonazza S, Turkington HL, Sukumar S, Peate E, Coutts HL, Montgomery JJ, Hawthorn C, Getty EMPE, Touzelet O, Barabas J, Power UF, **Courtney DG** (2026). Myoferlin is a component of late-stage vRNP trafficking vesicles for enveloped RNA viruses. *Nature Communications* 17. Cited 1.
2. Bonazza S, **Courtney DG** (2025). Influenza A virus RNA localisation and the interceding trafficking pathways of the host cell. *PLoS Pathogens* 21(4): e1013090. Cited 10.
3. Bonazza S, Coutts HL, Sukumar S, Turkington HL, **Courtney DG** (2024). Identifying cellular RNA-binding proteins during infection uncovers a role for MKRN2 in influenza mRNA trafficking. *PLoS Pathogens* 20(5): e1012231. Cited 11.
4. Dupont M, Krischuns T, Giai-Gianetto Q, Paisant S, Bonazza S, Brault J, Douché T, Perez-Perri JI, Hentze MW, Cusack S, Matondo M, Isel C, **Courtney DG***, Naffakh N# (2024). The RBPome of influenza A virus mRNA reveals a role for TDP-43 in viral replication. *Nucleic Acids Research*. Cited 18.
5. Coutts LH, **Courtney DG** (2024). Identifying individual pseudouridine (Ψ) sites across transcripts from HIV-1 infected cells. *HIV Protocols. Methods in Molecular Biology*, vol 2807.
6. **Courtney DG**, Tsai K, Bogerd HP, et al. (2019). Epitranscriptomic regulation of HIV-1 gene expression by m5C. *Cell Host & Microbe* 26(2): 217–227. Cited 261.
7. **Courtney DG**, Chalem A, Bogerd HP, et al. (2019). Extensive epitranscriptomic methylation of A and C residues on murine leukemia virus transcripts enhances viral gene expression. *mBio* 10(3): e01209-19. Cited 79.
8. **Courtney DG**, Kennedy EM, Dumm RE, et al. (2017). Epitranscriptomic enhancement of influenza A virus gene expression and replication. *Cell Host & Microbe* 22(3): 377–386. Cited 250.
9. **Courtney DG**, Moore JE, Atkinson SD, et al. (2015). CRISPR/Cas9 DNA cleavage at SNP-derived PAM enables both in vitro and in vivo KRT12 mutation-specific targeting. *Gene Therapy* 23(1): 108–112. Cited 129.
10. **Courtney DG**, Toftgaard Poulsen E, Kennedy S, et al. (2015). Protein composition of TGFBI-R124C- and TGFBI-R555W-associated aggregates suggests multiple mechanisms leading to lattice and granular corneal dystrophy. *Investigative Ophthalmology & Visual Science* 56: 4653–4661. Cited 39.

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